

Ratio test can't handle everything - 5.6

Example: ATCO $\sum_{n=1}^{\infty} \frac{3n^2+1}{10n^3-6}$.

(Using the LCT, we see $\lim_{n \rightarrow \infty} \frac{\frac{3n^2+1}{10n^3-6}}{\frac{1}{n}} = \frac{3}{10}$,

so this series diverges because $\sum_{n=1}^{\infty} \frac{1}{n}$ does.)

Ratio does not work here!

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{3(n+1)^2+1}{10(n+1)^3-6}}{\frac{3n^2+1}{10n^3-6}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{3(n+1)^2+1}{10(n+1)^3-6} \cdot \frac{10n^3-6}{3n^2+1} \right| = \lim_{n \rightarrow \infty} \left| \frac{30n^5 + \text{stuff}}{30n^5 + \text{other stuff}} \right| = 1$$

so the test is inconclusive.

In particular, the ratio test sees no difference between all the different p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} : \lim_{n \rightarrow \infty} \left| \frac{\frac{1}{(n+1)^p}}{\frac{1}{n^p}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n^p}{(n+1)^p} \right|$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^p = 1^p = 1$$

again ratio test is inconclusive.